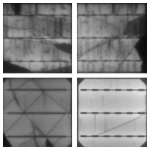
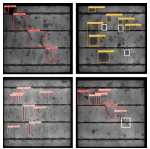
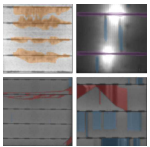
| Task Description | Data | | | | Model | | | |
|---|---|---|---|---|---|---|---|---|
| | Sample | Size | Annotations | Source | Algorithm | Architecture | Performance | Ref |
| PV Cells Classification |  | 2,624 samples<br>300 x 300 x 1 | 4 Classes of Pseudo Degradation Level | Openly Available [53] | DIP + ML Supervised | KAZE + SVM | F1 = 82.52% | [158] |
| | | | | | Supervised DL | CNN VGG19 | F1 = 88.39% | [158] |
| PV Cells Defects Detection |  | 36,543 samples<br>1024 x 1024 x 1 | 37,380 bounding box<br>4 Classes | Available Upon Request [159] | Supervised DL | YOLOv5l | mAP50 = 81.93% | [160] |
| | | | | | Supervised DL | ELCN-YOLOv7 | mAP50 = 91.93% | [161] |
| PV Cells Defects Segmentation |  | 17,064 samples<br>165 x 165<br>250 x 250<br>400 x 400 | 9 Classes of Anomalies | Model + Data Openly Available [162] | Supervised DL | DeepLabv3 w/ ResNet-50 | F1 = 69.6% | [152] |
| | | | | | Semi- -Supervised DL | Deeplabv3+ | F1 = 84.24% | [163] |